

Repair 1 - Workbench, grain, keys and NULL

Scope: SF0-SF1

- Connection problem -> verify server/service/credentials before changing SQL.
- Wrong database -> run/activate the controlled course schema.
- Repeated foreign key is not automatically duplicate data; restate row grain.
- NULL requires IS NULL / IS NOT NULL; compare COUNT(*) and COUNT(nullable_column).

Repair rule

- Keep the failed query/evidence.
- Repair only the weak idea.
- Close answer material before retry.
- Type the changed query from a blank tab and validate it.

Repair 2 - SELECT, WHERE, ORDER BY and LIMIT

Scope: SF2-SF4

- State desired result grain before DISTINCT.
- Write AND/OR logic in plain language and add parentheses deliberately.
- Top-N requires ORDER BY on the ranking field plus LIMIT.
- Fresh retry: changed filters + deterministic top five + boundary validation.

Repair rule

- Keep the failed query/evidence.
- Repair only the weak idea.
- Close answer material before retry.
- Type the changed query from a blank tab and validate it.

Repair 3 - Aggregates, GROUP BY and HAVING

Scope: SF5-SF7

- Decide whether the question needs one total or one row per group.
- Every selected non-aggregate field should fit the intended grouped grain.
- WHERE filters rows before grouping; HAVING filters groups after aggregates exist.
- Reconcile group counts/totals to a whole-table control.

Repair rule

- Keep the failed query/evidence.
- Repair only the weak idea.
- Close answer material before retry.
- Type the changed query from a blank tab and validate it.

Repair 4 - CASE and Gate transfer

Scope: SF8 / Gate

- Put the narrowest/highest-priority CASE conditions before broader ones.
- Always test exact boundaries and an ELSE path.
- Gate remains single-table: do not use JOIN/CTE/window syntax to solve around foundation reasoning.
- B/C are only explicit remediation retests after the prior Gate fails.

Repair rule

- Keep the failed query/evidence.
- Repair only the weak idea.
- Close answer material before retry.
- Type the changed query from a blank tab and validate it.